

INVITED TOPICAL REVIEW

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PREVENT Equations: Implications for Stroke Prevention

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ABSTRACT: Stroke remains a leading cause of death and long-term disability. Yet, much of its burden is preventable through earlier and more intensive management of vascular and cardiovascular-kidney-metabolic risk factors. Primary prevention is challenging as the first clinical event is often unpredictable and may manifest as stroke, myocardial infarction, heart failure, or peripheral arterial disease. Contemporary tools, therefore, estimate overall cardiovascular risk rather than stroke risk in isolation. The goal of this review is to emphasize that effective stroke prevention requires a shift toward global cardiovascular risk assessment and to highlight the potential clinical utility of newer risk prediction tools. The predicting risk of cardiovascular disease events equations, developed by the American Heart Association, update absolute risk estimation by modeling overall cardiovascular disease risk using sex-specific, race-free equations that incorporate kidney function, account for competing noncardiovascular death, and optionally include neighborhood deprivation. External validations suggest improved calibration compared with pooled cohort equations, supporting a more reliable estimation of absolute treatment benefit. For clinicians managing patients at risk for stroke, predicting risk of cardiovascular disease events is most useful when the clinical question is how aggressively to optimize risk factors before the first event, including decisions about blood pressure, lipid-lowering therapy, cardiovascular-kidney-metabolic management, and patient communication using 10-year, 30-year, and risk age outputs. Key limitations include its focus on overall rather than cause-specific cardiovascular risk, lack of stroke mechanism-specific estimates, absence of major nonatherosclerotic stroke drivers such as atrial fibrillation, and lack of imaging-stratified treatment-effect estimates. Predicting the risk of cardiovascular disease events' ultimate clinical impact and equity will depend on implementation within electronic health record workflows, autopopulated inputs, treatment pathways tied to estimated risk, and complementary cause-specific evaluation when clinically indicated.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: cardiovascular diseases ■ functional status ■ ischemic stroke ■ longevity ■ risk factors

Stroke remains a leading cause of death and long-term disability in the United States, annually affecting nearly 800 000 individuals and contributing to substantial morbidity and mortality.^{1–4} Beyond acute events, stroke accelerates vascular cognitive impairment, functional dependence, and caregiver and healthcare system burden across the lifespan.^{3,5} A central challenge to primary stroke prevention is that clinicians cannot reliably anticipate which atherosclerotic cardiovascular disease (ASCVD) manifestation, namely, ischemic stroke, myocardial infarction, peripheral arterial disease, or heart failure (HF), will occur first.⁶ Risk prediction tools,

therefore, appropriately target total cardiovascular risk, with atherosclerotic stroke treated as one component of broader ASCVD risk, rather than an isolated outcome.⁷

Cardiovascular-kidney-metabolic (CKM) health provides a unifying framework for this approach, hypertension, diabetes, chronic kidney disease (CKD), and obesity cluster, and acts synergistically to promote vascular injury, inflammation, and thrombosis, thereby increasing lifetime risks of ASCVD events and cognitive decline.^{8–14} From a brain-health perspective, strategies that address these shared CKM pathways can yield concurrent benefits across vascular outcomes, including stroke.^{8–10,12–14}

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The epidemiology of stroke also continues to evolve. While age-standardized stroke incidence has declined in some high-income settings, recent US data suggest plateauing and, in some groups, rising incidence, particularly among younger adults.^{15–17} Stroke burden is unevenly distributed: the southeastern Stroke Belt experiences some of the highest rates, and Black, Hispanic, and some Asian American subgroups bear disproportionate risks and poorer outcomes.^{18,19} These inequities reflect structural drivers, socioeconomic gradients, and neighborhood-level determinants of health.^{8,20} Accordingly, contemporary prevention frameworks increasingly incorporate neighborhood-level measures (eg, the social deprivation index [SDI]) to better reflect daily social and neighborhood conditions that shape vascular risk.^{8,21,22} Yet, despite established prevention frameworks, such as the American Heart Association (AHA) Life's Essential 8, a substantial gap persists between guideline recommendations and routine care risk-factor control, with major cardiovascular risk factors remaining suboptimally controlled at the population level.^{2,3,23} Closing this gap through earlier identification of high-risk individuals and intensification of evidence-based therapies could prevent a meaningful share of first-ever ASCVD events, including atherosclerotic stroke.^{3,5}

Reflecting this need for an earlier and more equitable risk assessment in primary prevention, the predicting risk of cardiovascular disease events (PREVENT) equations

have been incorporated into the 2025 hypertension and 2026 dyslipidemia guidelines as the preferred risk estimator to inform treatment thresholds and goals.^{24,25} Available through the AHA online calculator and integrated into mobile clinical decision-support tools, PREVENT equations provide estimated absolute risk at 10- and 30-year horizons (Table 1).^{11,26} For clinicians caring for patients at risk for stroke, PREVENT equations are most relevant when engaged upstream in primary prevention, where aggressive risk-factor optimization is warranted.^{2,11,13} Although stroke is included among PREVENT outcomes, the equations estimate global cardiovascular risk rather than mechanism-specific stroke risk. Thus, PREVENT equations do not separately estimate risk by stroke cause, such as cardioembolic stroke related to atrial fibrillation (AF), large-artery atherosclerosis, small vessel disease, or other non-atherosclerotic pathways.¹¹ This limitation underscores the need to combine global ASCVD risk estimation with stroke-specific risk assessment when clinically indicated.^{8,11} This review summarizes PREVENT equations' clinical utility for clinicians caring for patients at risk of stroke, its strengths and limitations, and practical applications for improving prevention equitably across large populations.



ADVANCING STROKE PREVENTION THROUGH RISK ESTIMATION

80% to 90% of stroke cases are preventable.^{27,28} Behavioral contributors such as smoking, unhealthy eating patterns, and physical inactivity remain major population-level targets for risk reduction.^{2,29} At the individual level, estimating absolute risk by integrating lifestyle, behavioral, and biological factors (eg, blood pressure [BP], lipids, glycemia, CKD, obesity, and sleep) enables precision primary prevention by identifying who benefits from earlier and more intensive intervention. Risk prediction converts silent risk into actionable estimates, enabling timely lifestyle and pharmacological interventions and appropriate treatment intensity based on expected benefit.^{30,31} Contemporary risk tools estimate overall ASCVD risk rather than stroke risk in isolation, as all ASCVD events share common upstream drivers. Within this framework, including stroke in the composite outcome improves the accuracy of predicted absolute risk and expected benefit from treatment, as emphasized in the AHA statement on incorporating stroke into cardiovascular risk prediction instruments.¹²

Accurate estimation of absolute cardiovascular risk is the first step at the point of care. It helps clinicians determine who should receive intensive lifestyle counseling, which medications to initiate, and how aggressively to treat BP, lipids, weight, and glycemia.⁸ It also clarifies the expected absolute risk reduction and number needed to treat. Because relative risk reductions for BP-lowering, statin, and CKM protective therapies are often broadly similar across baseline risk strata, the baseline absolute risk largely determines the absolute benefit, making calibration (getting the

Table 1. PREVENT Equations at a Glance: What It Predicts, What It Uses, and What It Outputs

Domain	Key points
Who	Adults aged 30–79 y without prior clinical CVD (primary prevention)
What it predicts	Total CVD, ASCVD (including stroke), and HF account for competing non-CVD deaths.
When	10 (age, 30–79) y; 30 (age, 30–59) y
Design	Sex-specific, race-free; intended for contemporary treated populations; and EHR-friendly inputs
Core inputs	Age, SBP+BP treatment, smoking, diabetes, non-HDL-C+HDL-C, statin use, and eGFR
Optional context	SDI (place-based); where available: albuminuria/HbA1c (implementation-dependent)
Outputs	Absolute risk (10/30 y) for total CVD/ASCVD/HF; risk age
Best use (stroke)	Guides intensity of prevention (BP, lipids, and CKM therapy), especially in stage-1 HTN and CKM-high-risk; imaging findings act as risk amplifiers near thresholds.
Not designed for	Mechanism-specific stroke prediction, such as AF-related cardioembolic, large-artery, or small vessel pathways; no imaging-stratified treatment-effect estimates
Guardrail	Use PREVENT for global ASCVD risk; run parallel mechanism-directed pathways when indicated.

AF indicates atrial fibrillation; ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; CKM, cardiovascular-kidney-metabolic; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; EHR, electronic health record; HDL-C, high-density lipoprotein-cholesterol; HF, heart failure; HTN, hypertension; PREVENT, predicting risk of cardiovascular disease events; SBP, systolic blood pressure; and SDI, social deprivation index.

absolute risk right) as crucial as discrimination (the ability to distinguish higher-risk from lower-risk patients).^{8,23}

Furthermore, lifetime and longer-term risk estimation is particularly important for stroke prevention.⁸ Younger or middle-aged adults with adverse CKM profiles may have low short-term risk but high cumulative lifetime risk, especially when hypertension, diabetes, CKD, obesity, or smoking are present. Longer-term estimates can, hence, support earlier risk communication, strengthen adherence to lifestyle and pharmacological prevention, and justify earlier intervention before vascular injury becomes clinically manifest.^{8,11,29} This is especially relevant for stroke because the burden of cumulative vascular exposure across midlife strongly influences later-life cerebrovascular disease, cognitive impairment, and disability.

For clinicians caring for patients at risk for stroke, the practical relevance of global risk estimation is greatest in high-risk individuals with signs of vascular disease, including asymptomatic carotid atherosclerosis, incidentally discovered covert brain infarction or extensive white matter disease, high-risk CKM profiles (eg, CKD with diabetes or obesity), a strong family history of premature ASCVD, or care delivered through multidisciplinary prevention clinics.^{2,3,8,12} In these contexts, a well-calibrated global risk estimate can support risk communication, guide intensification of evidence-based therapy, and harmonize recommendations across specialties.⁸ Secondary prevention remains central to vascular neurology, but primary-prevention risk tools are designed to identify and treat these high-risk individuals before the first ASCVD event.²³

Equally important is recognizing that stroke risk exists within the broader CKM continuum.^{8–10} Diabetes, CKD, obesity, and HF are not peripheral modifiers but major drivers of vascular risk. Furthermore, a CKM-anchored view improves calibration in certain risk groups, particularly CKD, which independently elevates stroke and broader brain-health risk.^{9,10} Readily available measures, such as estimated glomerular filtration rate and albuminuria, capture this kidney-brain axis and should inform absolute risk estimation and treatment thresholds.^{9,32}

Finally, the impact of PREVENT equations in contemporary practice will depend on both usability and fairness. Clinically useful tools should be Web-accessible, race-free, able to account for neighborhood context, and incorporate additional factors in the future (eg, inflammation, sleep, and social risk) as the evidence base matures. When these features are in place, global risk stratification becomes a practical and equitable tool to embed ASCVD and stroke prevention into routine clinical care.

EVOLUTION OF CARDIOVASCULAR RISK ESTIMATION

The Framingham Stroke Risk Profile and ASCVD pooled cohort equations (PCEs) were landmark multivariable

models that reshaped prevention by combining readily measured factors (age, BP, lipids, diabetes, smoking, and AF in the Framingham Stroke Risk Profile).^{7,33–35} In primary prevention, stroke is often considered within a broader ASCVD risk framework, and these global ASCVD risk models may provide stroke risk estimates that are similar in overall performance to stroke-specific algorithms in certain populations. However, both have important calibration and subgroup performance limitations in contemporary settings due to shifts in population risk profiles and changing demographics.^{12,36,37}

For clinicians taking care of patients at risk of stroke, the key shortcomings of these legacy calculators are less that they ignore stroke and more that they (1) may not perform well when applied to today's treated, diverse populations, and (2) incompletely capture contemporary CKM risk, particularly CKD.⁸ Both Framingham Stroke Risk Profile and PCEs are largely models based on one-time baseline measurements over a 10-year horizon, which heavily weight age and do not fully account for the possibility of dying from noncardiovascular causes before having a cardiovascular event. This can understate longer-term risk in younger adults with high-risk CKM profiles (eg, diabetes or CKD) while inflating short-term risk in older adults with multiple chronic conditions.^{7,33,34} The populations used to build the models and settings in which the models were tested also constrain generalizability (Framingham Stroke Risk Profile largely older White cohorts; PCEs primarily non-Hispanic White and Black adults aged 40 to 79 years), and external validations demonstrate overestimation/underestimation of actual risk when these legacy models are applied to modern health-system populations identified through electronic health records, with different treatment patterns and event rates.^{33,34,36–40} These limitations are clinically relevant because risk-based decisions can alter prevention intensity; more accurate risk classification can help identify patients who warrant earlier or more intensive lipid-lowering, BP control, and CKM-directed therapy, with the goal of reducing future ASCVD and cerebrovascular events.

In addition, traditional risk scores do not fully incorporate CKM drivers and contemporary biomarkers, which are now central to prevention decision-making.⁸ Key measures such as kidney function and injury (eg, estimated glomerular filtration rate and albuminuria), adiposity, and non-HDL-C (high-density lipoprotein-cholesterol)-based lipid risk are either absent or inconsistently incorporated across ASCVD and stroke prediction models.⁸ Notably, adiposity is not uniformly included across ASCVD/stroke prediction tools and, when incorporated in newer frameworks, may contribute differently to different end points. Although stroke mechanism and subtype diagnosis are important for guiding secondary prevention, most primary-prevention risk estimators are not designed to distinguish atherosclerotic from cardioembolic or other

nonatherosclerotic stroke pathways. As a result, absolute risk prediction should complement, not replace, mechanism-directed evaluation when indicated (eg, AF detection and anticoagulation decisions).^{2,8,12,41} Social determinants and neighborhood context, which strongly shape stroke outcomes, are also not directly represented in most legacy tools, and race-based coefficients raise ongoing concerns about fairness, interpretability, and performance across different clinical settings.^{14,20,41,42}

DEVELOPMENT OF PREVENT EQUATIONS

To address limitations of legacy risk tools, including sub-optimal calibration in contemporary treated populations, limited representation of diverse groups, incomplete capture of CKM risk, and insufficient accounting for competing noncardiovascular death, the AHA developed PREVENT equations for adults aged 30 to 79 years without prior cardiovascular disease (CVD), using large contemporary pooled data sets.¹¹ PREVENT equations differ from the legacy risk tools in several important ways, as summarized in Table 2.

The original PREVENT equations study used a split-sample approach, with approximately half of the participants used for equations development and the remaining participants used for internal validation, rather than relying on an independent external validation cohort.¹¹ In total, the development and validation samples included ≈6.6 million participants from 46 data sets across all US census regions.¹¹ The equations were designed to reflect age-related risk and explicitly account for competing non-CVD deaths, aligning predicted absolute risk with contemporary epidemiology.¹¹ In the internal validation sample from the original development study, PREVENT

showed good discrimination for total CVD, with median C statistics of 0.794 in women and 0.757 in men, and good calibration, with calibration slopes of 1.03 in women and 0.94 in men; similar performance was observed for ASCVD- and HF-specific models.¹¹

PREVENT equations estimate absolute risk for incident total CVD and its major subtypes: atherosclerotic CVD (ASCVD; including coronary events and stroke) and HF.^{8,11} In clinical practice, PREVENT equations provide 10-year risk for ages 30 to 79 years and 30-year risk for ages 30 to 59 years, enabling short- and long-term risk communication.^{8,11} While PREVENT equations incorporate kidney measures as predictors, they are not kidney-outcome models; their end points are CVD events (ASCVD/HF/total CVD).^{8,11} Importantly, and consistent with prior calls for stroke as a core vascular outcome in risk prediction, PREVENT equations include stroke within broader ASCVD outcomes, making it more relevant to stroke prevention than coronary-focused legacy discussions, while still functioning as a global cardiovascular risk tool rather than a stroke-specific prediction model.^{11,41}

PREVENT equations are sex-specific, race-free, and designed for use with routinely available clinical inputs (Table 1), facilitating Web-based calculation and electronic health record (EHR) autopopulation at the point of care.¹¹ These additions are particularly relevant for patients with CKD, in whom legacy tools may, as mentioned previously, be miscalibrated.^{8,11,14,43,44} While the social deprivation index provides a practical neighborhood-level substitute for social context, its ZIP code granularity underscores the need for richer person-level social determinants of health data as EHR data mature.⁴⁵

External evaluations across US and international cohorts demonstrated that PREVENT equations are

Table 2. PREVENT Versus Legacy Risk Equations: Key Differences Relevant to Stroke Prevention

Feature	Legacy tools (typical)	PREVENT	Stroke prevention implication
Outcomes	Often, coronary-weighted composites and stroke are variably represented.	Models ASCVD (includes stroke) and HF within the total CVD framework	Improves stroke relevance while reflecting uncertainty in the first event (stroke/MI/HF)
Calibration/transportability	Derived in older cohorts and frequent miscalibration in contemporary treated populations	Developed/validated in large, diverse contemporary data sets, and calibration emphasized	More credible absolute risk for ARR/NNT-informed intensity decisions
Equity approach	Race coefficients common and interpretability/fairness concerns	Race-free, sex-specific; optional SDI	Supports scalable, equity-forward implementation; SDI may identify support needs.
CKM/kidney inputs	Kidney function often absent or indirect	eGFR included (CKM-aware)	Better risk estimation in CKD and other CKM-high-risk phenotypes
Competing risk	Non-CVD death is often not modeled.	Competing non-CVD death modeled	Helps contextualize net benefit in older, multimorbid adults
Time horizon/communication	Typically 10 y only	10+30 (30–59) y; risk age available	Enables life-course counseling, especially when 10-y risk appears low in younger adults
Mechanism specificity	Stroke-specific tools may include AF; ASCVD tools typically do not.	Not mechanism-specific; AF is not included as a predictor/input.	Requires parallel mechanism-directed pathways (eg, AF detection/anticoagulation frameworks)
Imaging	Imaging used as risk enhancers outside most calculators	Imaging not in base model; can act as risk amplifier alongside PREVENT.	Use imaging to intensify foundational prevention; avoid treat-the-scan defaults (aspirin/procedures).

ARR indicates absolute risk reduction; ASCVD, atherosclerotic cardiovascular disease; CKM, cardiovascular-kidney-metabolic; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; HF, heart failure; MI, myocardial infarction; NNT, number needed to treat; PREVENT, predicting risk of cardiovascular disease events; and SDI, social deprivation index.

often better calibrated than legacy risk tools (Supplemental Text).^{46–54} However, variation across populations and settings reinforces the need for local calibration and subgroup monitoring during implementation (Table S1). A recent extension of PREVENT risk age equations translates absolute risk into the age of an optimally healthy individual, allowing comparison of risk age with chronological age.²⁹ This may provide a more intuitive counseling tool than percentages alone, particularly for younger adults whose 10-year risk appears low despite substantial lifetime exposure.²⁹ The clinical inputs, modeled outcomes, time horizons, and key outputs of PREVENT most relevant to clinicians are summarized in Table 1.

ELEVATED STROKE RISK: APPLYING PREVENT TO STROKE PREVENTION

Risk prediction is the gateway to primary prevention, not an academic exercise (graphic abstract). Both the US Preventive Services Task Force and the AHA endorse absolute risk assessment to guide primary prevention.² PREVENT puts this into practice by linking risk estimates to treatment thresholds, quantifying expected benefit (absolute risk reduction and numbers needed to treat, often expressed as events prevented per 1000 treated), and, when helpful, using risk age to translate percentages into an intuitive communication tool.^{8,11,46,47} As relative treatment effects are generally similar across baseline risk, addressing several risk factors together

(BP, atherogenic lipids, and CKM drivers) yields greater absolute benefit than isolated, stepwise changes.⁸ This also highlights the value of team-based care models and multidisciplinary prevention.

For clinicians, the highest yield workflow employing PREVENT is (1) calculate risk with routinely available EHR data; (2) communicate risk using long-horizon estimates and risk age when short-term risk appears deceptively low; (3) act using grouped prevention strategies that reliably reduce stroke risk across populations (BP control, lipid lowering, CKM therapy optimization, and smoking cessation); and (4) overlay cause-specific evaluation and treatment pathways when cardioembolic or other nonatherosclerotic mechanisms are plausible (eg, AF detection and anticoagulation decisions).⁵⁵ Table 3 summarizes a pragmatic PREVENT-to-action map and emphasizes important guardrails and cautions to avoid substituting global risk estimation for mechanism-specific evaluation.

THERAPEUTIC IMPLICATIONS

Blood Pressure

Hypertension is the single most powerful modifiable risk factor for first-ever stroke, with risk rising continuously across the BP spectrum.⁵⁶ PREVENT operationalizes guideline-concordant escalation by identifying patients in whom earlier pharmacotherapy and tighter control yield the greatest absolute benefit, particularly

Table 3. Translating Prevent Risk Into Action: What to Do Next (Stroke Prevention-Focused)

Domain	When PREVENT is most useful	What to do next (action)	Stroke-facing guardrail
Blood pressure	Stage-1 HTN where treatment depends on global risk	Use PREVENT 10-y total CVD risk to guide earlier pharmacotherapy and intensification; target <130/80 with home/ambulatory confirmation.	BP is a dominant modifiable driver of ischemic and hemorrhagic stroke; PREVENT informs timing/intensity, not whether to treat.
Lipids	Borderline/intermediate risk and CKM-high-risk phenotypes	Anchor statin intensity to absolute risk/benefit; use risk enhancers when decisions are close; and escalate therapy when residual LDL/non-HDL risk remains high.	PREVENT may yield lower 10-y risk than legacy tools; avoid undertreatment by emphasizing ARR/NNT, long-horizon risk, and risk age.
CKM optimization (diabetes/obesity/CKD)	Diabetes, obesity, and CKD, where life-course risk is underappreciated	Intensify therapies with proven CV benefit, address obesity upstream, and integrate kidney-protective strategies where indicated.	PREVENT captures much of obesity via downstream factors; it still treats adiposity directly. CKD is a major stroke risk amplifier; respond with durable control.
Smoking/nicotine	Any predicted risk	Treat cessation as universal high-yield therapy; combine counseling+pharmacotherapy, and connect to resources	Do not wait for a high score; cessation is always indicated.
Antithrombotics (aspirin)	Patients/clinicians seeking aspirin based on high risk	Avoid routine aspirin for primary prevention; individualize only in selected very-high-thrombotic/low-bleeding-risk settings.	PREVENT provides context but should not drive aspirin initiation absent a clear indication.
Mechanism-directed pathway	When cardioembolism/non atherosclerotic causes are plausible	Run parallel pathways (eg, AF detection/monitoring and anticoagulation decisions per AF frameworks).	PREVENT is not mechanism-specific; do not substitute global risk for etiologic evaluation.
Imaging as a risk amplifier (carotid plaque, WMH/CSVD)	When findings are incidental or decisions are near thresholds	Use imaging to reinforce/intensify foundational prevention (BP, lipids, CKM, and smoking)	Avoid treat-the-scan reflexes; PREVENT does not provide imaging-stratified treatment-effect estimates.

AF indicates atrial fibrillation; BP, blood pressure; CKD, chronic kidney disease; CKM, cardiovascular-kidney-metabolic; CSVD, cerebral small vessel disease; CV, cardiovascular; CVD, cardiovascular disease; HDL, high-density lipoprotein; HTN, hypertension; LDL, low-density lipoprotein; PREVENT, predicting risk of cardiovascular disease events; and WMH, white matter hyperintensity.

those with stage-1 hypertension and elevated predicted risk and those with CKM burden (eg, CKD and diabetes), where absolute risk is often underappreciated (Table 3).^{2,24,57,58} In a recent secondary analysis of the Systolic Blood Pressure Intervention Trial participants without baseline CVD, Derington et al⁵⁹ showed that the relative benefit of intensive versus standard systolic BP treatment was broadly consistent across PREVENT-defined risk strata, whereas the absolute benefit increased with higher PREVENT-predicted risk; the 4-year absolute risk difference for the primary CVD outcome increased from 0.2% in the low-/borderline-risk group to 1.5% and 2.4% in the intermediate- and high-risk groups, respectively.⁵⁹ These findings support the use of PREVENT equations to match BP treatment intensity to absolute CVD risk, rather than relying solely on BP levels.

Lipids

For lipids, PREVENT supports a discussion focused on expected clinical benefit by estimating 10- and 30-year ASCVD risk and helping clinicians match lipid-lowering intensity to absolute risk.¹¹ This approach is now reflected in the 2026 AHA multisociety dyslipidemia guidelines, which incorporate PREVENT for primary-prevention lipid-lowering decisions and reintroduce LDL-C (low-density lipoprotein-cholesterol) and non-HDL-C treatment goals.²⁵ In adults without ASCVD and LDL-C of 70 to 189 mg/dL, the guideline uses PREVENT-ASCVD risk to support earlier lipid-lowering therapy, with LDL-C goals of <100 mg/dL for low-, borderline-, or intermediate-risk adults and <70 mg/dL for high-risk adults.²⁵ PREVENT is particularly helpful in CKM-high-risk profiles, where CKD, diabetes, and adiposity can amplify vascular risk; when treatment decisions remain uncertain, risk-enhancing information, such as Lp(a; lipoprotein[a]), apoB, and selective coronary artery calcium scoring, can further refine the decision to initiate or intensify lipid-lowering therapy.^{2,8,25,60} Table 3 summarizes how to anchor statin intensity and adjunct therapies to the expected absolute benefit while using risk enhancers when decisions remain borderline.

Diabetes and Obesity

Diabetes and obesity accelerate stroke risk through the CKM continuum.^{2,10} PREVENT helps shift management beyond glucose control alone toward overall vascular risk reduction, supporting earlier intensification of therapies with proven cardiovascular benefit and better alignment of counseling with the patient's lifetime risk (30-year risk and risk age).¹¹ Because obesity's contribution is often mediated through downstream predictors (BP, diabetes, and lipids), clinicians should still treat adiposity directly

with structured lifestyle, pharmacotherapy, and, when appropriate, metabolic surgery while using PREVENT outputs to communicate cumulative risk and motivate adherence (Table 2).^{8,61,62}

Antithrombotic Therapy

Antithrombotic decisions in primary prevention remain based on the clinical indication and likely cause of stroke.^{2,35} PREVENT can contextualize overall vascular risk, but it should not be used to justify routine aspirin use in the absence of a clear indication. In a national health and nutrition examination survey analysis of adults aged 40 to 59 years without ASCVD or major bleeding-risk factors, Sayed et al⁶³ found that applying a 10-year ASCVD risk threshold of $\geq 10\%$ reduced aspirin eligibility from 8.3% using the PCEs to 1.2% using PREVENT, underscoring that PREVENT-specific treatment thresholds may be needed rather than directly substituting PREVENT into older PCE-based recommendations. The practical role for clinicians is to maintain a focus on foundational risk-factor control while triggering mechanism-directed pathways (eg, AF evaluation and anticoagulation frameworks) when appropriate (Table 2).^{2,11,55}

INNOVATIONS IN STROKE RISK PREDICTION AND IMPLEMENTATION

Extracranial Carotid Atherosclerosis (Asymptomatic)

Routine population screening for asymptomatic carotid stenosis is not recommended (class III, no benefit, level B-NR).² Although cervical bruit has long been recognized as a marker of increased vascular risk, targeted carotid duplex may be reasonable only in selected patients when results would change management, such as patients with a bruit plus additional atherosclerotic risk features, known systemic atherosclerosis, or clinical circumstances in which defining carotid severity would prompt closer surveillance, referral, or procedural discussions.^{2,64–66} When severe asymptomatic carotid stenosis ($>70\%$) is detected, management should be individualized via shared decision-making that weighs perioperative risk, plaque/anatomic features, life expectancy, and patient preferences.² In all cases, optimized medical therapy is foundational, including intensive risk-factor modification (BP, lipids, smoking, and lifestyle) and, in most patients with established carotid atherosclerosis, single-antiplatelet therapy as part of broader atherosclerotic risk management rather than routine aspirin for everyone.² Recent CREST-2 results underscore that procedural decisions for asymptomatic carotid stenosis should be individualized and interpreted in the context of contemporary

intensive medical therapy.⁶⁷ PREVENT complements this pathway by quantifying 10- and 30-year total CVD and stroke risk within a CKM framework, providing the baseline absolute risk estimate (from clinical/biomarker inputs), while carotid imaging can function as a risk modifier when decisions are near treatment thresholds, when procedural referral is being considered, or when added convincing is needed for adherence to intensive prevention.¹¹ Finally, as Dolmans et al⁶⁸ highlight, using the explicit diagnostic criteria for transient ischemic attack in settings where mimics (eg, migraine) are common, vascular imaging findings alone should not dictate care; structured clinical tools plus CKM context can help avoid both overtreatment and undertreatment.

Asymptomatic Cerebral Small Vessel Disease: Silent Infarcts and White Matter Hyperintensities

Silent infarcts and white matter hyperintensities are common ($\approx 10\%$ – 20% of adults) and predict future stroke, dementia, and accelerated neurodegeneration.^{69,70} First-line therapy is aggressive risk-factor control (class I, C-LD).² Statins may be considered in adults with silent infarcts who do not otherwise meet lipid thresholds (class IIb, B-NR), with observational data suggesting slower cerebral small vessel disease progression and fewer strokes among statin users (HR ≈ 0.56 – 0.60).² Aspirin is not recommended routinely in cerebral small vessel disease as an incidental imaging finding without another clear atherosclerotic indication, given inconclusive benefit (class IIb, C-LD).^{2,71} In practice, antiplatelets are often used when concomitant atherosclerotic disease is present (eg, CAD/peripheral arterial disease/carotid disease) or when thrombotic risk is high, and bleeding risk is low, but this is driven by the broader vascular indication, not cerebral small vessel disease alone. PREVENT complements this approach by quantifying 10- and 30-year total CVD risk (including stroke) within a CKM framework, strengthening the rationale to intensify BP, lipid, glyce-mic, kidney, and weight-directed therapies and discouraging a reflexive treat-the-scan strategy.^{11,72}

Imaging-Based Risk Refinement and the Role of PREVENT

Imaging markers, such as carotid plaque or stenosis and cerebral small vessel disease features, may serve as risk amplifiers, but routine imaging as a universal component of PREVENT-based risk assessment is not established and is not currently recommended.¹¹ These findings should not be interpreted as automatic triggers for procedures or medications. Per guidelines, the foundational response remains class I personalized prevention and

risk-factor optimization.² PREVENT can contextualize these findings by providing calibrated 10- and 30-year estimates of total CVD and ASCVD risk from clinical and biomarker inputs.^{11,46,47} However, PREVENT does not yet provide lesion-specific, imaging-stratified estimates of treatment effect (eg, aspirin net benefit or revascularization thresholds), and this evidence gap should temper interpretation and downstream decision-making.

Consistent with guideline calls to integrate imaging with risk estimation, future work linking imaging phenotypes to PREVENT-predicted risk, alongside bleeding-risk models, could help refine thresholds and better identify subgroups most likely to benefit from escalated therapies.^{2,8} Echoing Explicit Diagnostic Criteria for Transient Ischemic Attack validation, structured tools should inform, but never replace, clinical judgment when applying imaging findings to prevention decisions.⁶⁸

INTEGRATION INTO CLINICAL PRACTICE: IMPLEMENTATION CONSIDERATIONS

Stroke prevention ultimately depends on whether risk assessment is embedded in routine care, where most vascular events are prevented through sustained control of BP, lipids, diabetes/obesity, and smoking.¹¹ The 2024 AHA/American Stroke Association guideline elevates standardized screening of traditional risk factors and social determinants of health and places routine risk estimation at the center of primary prevention.² PREVENT is well-suited for EHR deployment because its core inputs (age, sex, systolic BP/antihypertensive use, lipids, diabetes, smoking, and estimated glomerular filtration rate) are structured and commonly available, with optional add-ons (HbA1c, albuminuria, and SDI) that capture CKM burden and structural disadvantage.¹¹ Autopopulation makes calculation universal, easy, seamless, and scalable across clinical settings.⁷³

For clinicians, the PREVENT equations should be framed as global cardiovascular risk estimators that support intensity-matched vascular prevention, while cardio-embolic prevention remains driven by rhythm surveillance and anticoagulation pathways.¹¹ Implemented this way, EHR decision support can pair PREVENT outputs with stroke-relevant actions such as BP and statin intensification cues, preloaded orders/smartphrases for shared decisions, and prompts for mechanism-focused pathways when appropriate (eg, AF screening).^{8,11}

PREVENT-based panels/registries can identify patients with high predicted vascular risk who are undertreated and support proactive outreach and protocolized titration.^{8,74} Because PREVENT replaces race with place-based SDI, it can reduce misclassification and help connect patients to adherence-enabling supports (such as medication affordability programs, transportation assistance, and food security resources) that materially affect stroke

risk.⁸ Clinician education should be brief, embedded, and practical: interpreting 10- and 30-year estimates, communicating absolute risk and benefits, executing CKM-first intensification, and applying equity-informed counseling using SDI. These skills can be reinforced through micro-learning, templated notes, and audit-and-feedback to improve consistency while minimizing alert fatigue.⁷⁴

FUTURE DIRECTIONS: PRECISION STROKE PREVENTION

Future work should focus on using PREVENT as a pragmatic foundation for earlier and more equitable stroke prevention while testing whether additional data sources improve clinical decision-making beyond routinely available variables.^{2,8} Because stroke risk changes over time with aging, BP control, kidney function, glycemia, medication adherence, incident AF, and evolving vascular disease, implementation studies should evaluate dynamic PREVENT-based workflows that update risk estimates as clinical data change and prompt timely treatment intensification.²

Artificial intelligence, digital health data, imaging, biomarkers, genetics, and multiomics approaches may eventually refine risk assessment in selected patients, but they should be incorporated only when they improve calibration, clinically meaningful reclassification, treatment decisions, or equity.^{75–78} Potential high-yield settings include younger adults with high lifetime but low 10-year risk, patients with a premature family history or discordant risk phenotypes, and individuals in whom additional information would change surveillance or prevention strategies.^{77–79} Artificial intelligence-enabled tools may also help identify occult AF risk, lapses in medication adherence, or follow-up failures, but adoption should require transparent validation, monitoring for bias and drift, and clear integration into clinical workflows.^{76,80}

The guiding principle is that PREVENT should remain the accessible baseline for absolute CVD risk estimation, while advanced tools should be reserved for situations in which they measurably improve stroke-preventive care.

REIMAGINING STROKE PREVENTION IN THE PREVENT ERA

PREVENT marks an important shift in risk prediction by modeling ASCVD, including stroke, alongside HF, moving beyond models focused mainly on coronary disease and aligning prevention with the shared CKM substrate that drives cerebrovascular events.^{8,11} For stroke clinicians, the central opportunity is not to replace mechanism-specific evaluation but to use well-calibrated global risk estimation to identify high-risk individuals earlier, communicate risk more effectively, and match prevention intensity to expected absolute benefit.

Several priorities should guide the next phase of stroke prevention in the PREVENT era. First, future risk estimation should evolve toward models that incorporate stroke subtype and patient age. In younger patients, stroke mechanisms are proportionally more heterogeneous (eg, nonatherosclerotic arteriopathies and cardiac shunts) than in older adults, and prediction frameworks should reflect this diversity rather than assuming a uniform atherosclerotic pathway.^{2,12,16} Second, ongoing model updates should be treated as a requirement, not an aspiration: periodic recalibration and transparent reporting of subgroup performance and error metrics are essential to preserve trust and clinical utility.¹¹ Third, SDI should function as a bridge toward granular person-level social determinants of health capture as EHRs mature, enabling more precise targeting without reintroducing fairness concerns.⁴⁵ Fourth, PREVENT's performance should also be evaluated in groups in whom risk mechanisms may differ from typical ASCVD patterns, including people with chronic inflammatory conditions, cancer, and other high-risk comorbidities. Future iterations should also evaluate emerging biomarkers, sleep health, and clear treatment pathways tied to risk thresholds.

Implementation science will ultimately determine PREVENT's impact.⁷⁴ PREVENT should be embedded into EHR workflows with autopopulated inputs, risk-linked order sets, and concise documentation tools that translate risk into stroke-preventive actions.^{8,73} Risk tiers should map to pragmatic treatment bundles that focus on what most reliably reduces stroke risk across populations, BP optimization, lipid lowering, diabetes/obesity management, and smoking cessation while maintaining parallel pathways for mechanism-specific prevention (eg, rhythm surveillance and anticoagulation decision support when cardioembolism is plausible).^{9,10} Multidisciplinary teams will be essential: pharmacists can support medication titration and adherence, nurses can coordinate BP monitoring and follow-up, nutritionists can address diet and weight management, and community health workers can help overcome access barriers. Health systems could use PREVENT-based thresholds to create population-level registries to prioritize outreach and pharmacist-led medication titration.

Finally, equity and innovation must advance together. Validation in low-income and low-health literacy communities, attention to data missingness and interoperability, and mitigation of digital access barriers are central to preventing widening disparities. Remote monitoring, patient portals, and SMS outreach can help convert risk identification into between-visit action. In selected settings, artificial intelligence and multiomics may add incremental value but only within guardrails that prioritize calibration, clinical actionability, and equity.⁷⁴ A broader brain-health framing, using constructs such as risk age and pragmatic cognitive end points, may further unify prevention strategies across stroke and dementia and strengthen patient motivation without over-medicalization.^{74,81}

CONCLUSIONS

PREVENT modernizes primary-prevention risk assessment by estimating overall cardiovascular risk, including ASCVD events such as stroke, while also incorporating HF, CKM biology, kidney function, competing non-CVD death, and a sex-specific, race-free framework with optional SDI to support more equitable, clinically actionable estimates of absolute risk. For clinicians, PREVENT is most valuable when the clinical question is how aggressively to optimize vascular risk factors before the first event, rather than determining the stroke mechanism. When applied effectively, it enhances risk communication through 10- and 30-year risk estimates and risk age estimates, improves calibration in contemporary treated populations, and helps align the intensity of BP, lipid, and CKM-directed therapies with the patient's expected absolute benefit.

PREVENT's impact will hinge on effective implementation. Embedding PREVENT into EHR workflows with autopopulated inputs, risk-linked therapy bundles, and parallel pathways for mechanism-specific evaluation (eg, AF detection/anticoagulation when indicated) can translate risk estimation into sustained risk-factor control at scale. Future advances should prioritize ongoing recalibration, extensions that account for stroke subtype and age-specific mechanisms, granular person-level social determinants of health capture beyond ZIP code proxies, and carefully governed artificial intelligence or omics enhancements that improve calibration, actionability, and equity.

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Supplemental Material

Supplemental Text
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